RTSP A.I. Variables

Capabilities:

1. Under the correct circumstances, a 95% success rate with determining if an image is a human or vehicle

2. False Positive reduction rate of 95% or higher

3. Human or vehicle only detection (no bugs or trees will trigger alerts)

4. Increased distance detection capabilities under the correct circumstances when applying High Definition (HD)

5. Logo detection capabilities (enhanced feature set)

6. All alerts accessed through easy-to-use dashboard hosted by Centralized Vision (CV)

7. Offers Live View of the camera on the A.I. dashboard

8. Alerts include a video of the activity causing the alert

Requirements:

1. Each camera must be able to output at least two Real Time Streaming Protocol (RTSP) streams over TCP/IP (Transmission Control Protocol/Internet Protocol). One stream, configured as a low-resolution stream, must be 180pixels (p) or greater but lessor or equal to 480p (720 x 480), with a bitrate lessor or equal to 1 Megabits per second (Mbps). The second stream, configured as a high-resolution stream can be any resolution equal to or less than 2592 x 1944, which therefore includes 1080p HD, with a bitrate lessor or equal to 3 Mbps.

2. Daylight object detection - human and or vehicle, can be up to 50 feet / 15 meters. Night-time object detection - human and or vehicle is dependent on camera resolution, a clean lens, adequate ambient light, artificial light, camera Infrared (IR) performance and normal weather conditions.

3. Both RTSP streams must be encoded with H.264, and the encoding must not be multi-sliced. Other encodings like Motion JPEG and H.265 are not supported.

4. Unreliable connectionless transmission protocol (UDP) streams are not supported.

5. 180- and 360-degree cameras must be able to send preprocessed de-warped streams for proper viewing and A.I. analysis.

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RTSP A.I. Variables

Limitations:

Dark clothing prevents the A.I.'s ability to accurately determine if what's being analyzed is a human in poor light or poor camera conditions.

Persons attempting to blend in by hovering along or walking along a chain link fence will confuse A.I. and likely prevent it from determining if what's being analyzed is a human.

Cameras that are dirty/blurry will negatively affect the A.I.'s ability to accurately determine if what's being analyzed is a human or vehicle.

Object (human/vehicle) detection range is reduced and typically limited to 50 feet/15 meters due to the need for a Low-Resolution Stream to be utilized.

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SMTP A.I. Variables

Capabilities:

1. Under the correct circumstances, a 95% success rate with determining if an image is a human or vehicle.

2. False Positive reduction rate of 95% or higher.

3. Human or vehicle only detection (no bugs or trees will trigger alerts)

4. Increased distance detection capabilities under the correct circumstances when applying High Definition (HD).

5. Logo detection capabilities (enhanced feature set)

6. All alerts accessed through easy-to-use dashboard hosted by Centralized Vision (CV)

Requirements:

1. Each camera system must be able to send Simple Mail Transfer Protocol (SMTP) email alerts with images attached to the Artificial Intelligence (A.I.) cloud for analysis. (multiple snapshots/video clips recommended).

2. A Minimum resolution of 360p is required to ensure (accurate) detection, we however strongly recommend a resolution of 480p (720 x 480, 704 x 480 or 960 x 480) for best performance based on detection range. Detection range refers to the distance from the camera, a vehicle or human can be detected. Better resolution, better ability.

3. Standard A.I. (Standard Definition) - Object detection - human and or vehicle can be up to **50 Feet** or greater and is solely dependent on the quality of the camera, coupled with ideal lighting, a clean lens and normal weather conditions**.** Object detection/range can be extended to 140 feet by offering Light High Definition A.I. (HD) instead of SD. Camera requirements for HD are at minimum 720P X 480P and again dependent on the conditions above.

4. Low light conditions (usually happening at the edge of a Region of Interest - RIO) can affect detection capabilities, with false or missed alerts due to pixel differentiation issues.

5. Strong backlight/glare typically from a nearby spotlight, sunlight or even headlight and or near objects reflecting IR (infrared) will affect detection capabilities. If those issues are of concern, HD should be a strong consideration as the alternative.

6. Panorama, thermal cameras, and PTZ (pan/tilt,/zoom) cameras are not supported.

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SMTP A.I. Variables

Limitations:

A.I. detections are transmitted to CV via the local ISP (internet service provider) and via internet. Typical transmission times vary but may take up to 1 minute to arrive; in rare instances may take as long as 4-5 minutes to arrive.

A.I. does not typically sync and work with equipment cloud-based solutions - unless SMTP with image is available.

Dark clothing prevents the A.I.'s ability to accurately determine if what's being analyzed is a human in poor light or poor camera conditions.

Persons attempting to blend in by hovering along or walking along a chain link fence will confuse A.I. and likely prevent it from determining if what's being analyzed is a human.

Camera's that are dirty/blurry will negatively affect the A.I.'s ability to accurately determine if what's being analyzed is a human or vehicle.

Cameras that send single images per SMTP alert are less accurate due to limited data for A.I to review.